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21 October 1985
SUPERSEDING
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28 June 1978

MILITARY SPECIFICATION

LATHE, ENGINE, 9 BY 22 INCH CAPACITY, BENCH TYPE

This specification is approved for use within the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 9 by 22 inch capacity bench type, motor-driven engine lathe with quick change gears.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3416

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-681 - Identification Coding and Application of Hookup and Lead Wire.
MIL-STD-889 - Dissimilar Metals.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.8-72 - Chucks and Chuck Jaws.
ANSI B5.9-72 - Spindle Noses.
ANSI B5.10-72 - Machine Tapers.
ANSI B5.16-72 - Accuracy of Engine and Tool Room Lathes.

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRIC MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems.

(Applications for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code.
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

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3.2 Design. The lathe shall be new, shall be of the manufacturer's latest design meeting the performance requirements specified herein, and shall meet or exceed the accuracy requirements of ANSI B5.16-72. The lathe shall be a thread cutting, back geared, bench type with quick change gears. The design shall be such that required maintenance can be performed without using special tools. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair or replacement without modification, distortion, or damage to the machine or any machine component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the lathe. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

3.2.3 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the lathe to remove the temporary caps or stoppers before start-up or use of the lathe.

3.2.4 Gears. All gears and pinion designs selected for use in the lathe or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380-84.

3.2.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

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3.2.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The lathe shall comply with the standards promulgated under OSHA 2206 that are applicable to the lathe itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The engine lathe shall be constructed of parts and materials which are without defects. The lathe shall be furnished complete so that when installed and connected to a proper power source it may be used for all operations for which it was designed.

3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.4.5 Dials and plates. Markings on indicating and measuring dials and on instruction and name plates shall be permanent and shall be in plain, legible, boldface English language characters on a contrasting background.

3.4.6 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

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3.4.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied. The paint coating shall be not less than 2.0 mil in thickness. The paint adhesion shall be such that no paint shall be peeled by fiberglass tape when two intersecting cuts expose the base metal for a distance of not less than two inches. The surface hardness of the paint shall be between paint industry standards H to HB.

3.5 Performance and product characteristics

3.5.1 Performance. The lathe shall be capable of turning a piece of SAE 1020 or 1040 cold rolled steel, 1 1/2 inches in diameter and 20 inches long, to an approximate diameter of 1 7/16 inches for a distance of 18 inches with departure from a cylindrical surface not to exceed ±0.002 inch. The lathe shall be capable of turning a piece of SAE 1020 or 1040 cold rolled steel approximately five inches in diameter by four inches long to a diameter of 4 3/4 inches and facing-off the end. The faced surface shall be square with the cylindrical surface within 0.001 inch. Surface roughness shall not exceed 63 microinches, arithmetical average (aa).

3.5.2 Dimensions and characteristics. Unless otherwise specified (see 6.2.1), the lathe shall have the dimensions and characteristics specified in Table I.

3.6 Components.

3.6.1 Lathe bed. The lathe bed shall be structurally designed and built to resist distortion due to workloads. The top of the bed shall have precision finished ways (flat and V-ways) to afford alignment of the headstock, carriage, and tailstock.

3.6.2 Headstock.

3.6.2.1 Headstock housing. The headstock housing shall be of rigid construction and shall be fitted to the ways in such a manner that neither the headstock housing nor the bed frame will be distorted when the bolts are tightened.

3.6.2.2 Spindle. The spindle nose shall conform to applicable requirements of ANSI B5.9-72. Bearing and mount surfaces shall be hardened and ground to prevent wear.

3.6.2.3 Bearings. The headstock spindle bearings shall be either anti-friction ball or roller bearings or bronze sleeve type bearings. The bearings shall be capable of producing the spindle radial and axial movements within the accuracy requirements specified in ANSI B5.16-72.

3.6.3 Tailstock.

3.6.3.1 Tailstock housing. The tailstock housing shall not prevent the compound rest from being swiveled parallel to the bed. A clamping device shall be

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provided which shall be capable of locking the tailstock at any point along the lathe bed. The base of the tailstock shall be fitted to the ways in such a manner that tightening of the tailstock to the lathe bed shall not cause misalignment with the headstock.

TABLE I. Dimensions and characteristics.

Swing over bed and saddle wings (minimum)	9 inches
Distance between centers (minimum range)	0 to 22 inches
Spindle speed changes (minimum)	6 forward and 6 reverse
Low speed, revolutions per minute (rpm) (maximum)	55 or less (forward and reverse)
High speed, rpm, (minimum)	750 or more (forward and reverse)
Thread cutting range, number of pitches (minimum)	36 (right hand and left hand)
Range of threads per inch (including all standard inch system pitches listed in FED-STD-H28)	4 or less to 96 or more
Longitudinal feed range, inches per revolution	0.004 or less to 0.0128 or more
Cross feed range, inches per revolution	0.0016 or less to 0.012 or more
HEADSTOCK	
Spindle hole (minimum)	3/4 inch
Collet capacity (minimum)	1/16 to 1/2 inch
Headstock center taper per ANSI B5.10	As applicable to spindle nose
Spindle nose	1 1/2-8, D1 camlock, or Type LO
TAILSTOCK	
Spindle diameter (minimum)	1 inch
Spindle travel (minimum)	2 1/8 inches
Spindle taper per ANSI B5.10-72	No. 2
Center taper per ANSI B5.10-72	No. 2
Set-over (minimum) either way from center	1/4 inch
COMPOUND REST	
Cross slide travel (minimum)	5 1/4 inches
Angular feed travel (minimum)	2 inches

3.6.3.2 Tailstock set-over. The tailstock base shall be provided with a set-over adjustment, actuated by a screw, or screws, to permit turning of tapers between centers. A reference mark shall be provided for measuring the amount of set-over for taper turning.

3.6.3.3 Tailstock spindle. The tailstock spindle shall be precision ground and shall have marked graduations in increments not greater than 1/16 inch, progressive with extension. The spindle shall be threaded internally to receive a handwheel actuated spindle screw on one end. The other end shall be bored and ground to receive a standard Morse No. 2 taper center. A spindle binder shall be provided to enable locking of the spindle at any stage of extension without altering the alignment of headstock and tailstock centers more than $\pm$0.0005 inch. A means shall be provided for automatic ejection of the tailstock center when the spindle is retracted.

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3.6.3.4 Headstock-tailstock alignment. Vertical headstock-tailstock alignment shall be between zero and 0.001 inch high at the tailstock end when measured on a 12 inch long test mandrel. Horizontal headstock-tailstock alignment shall be within ±0.0005 inch when using the 12 inch long test mandrel.

3.6.4 Carriage.

3.6.4.1 Thread chasing dial indicator. A thread chasing dial indicator shall be provided and shall be located on the carriage. Dial markings shall indicate the correct engagement of the lead screw and half nut when cutting threads.

3.6.4.2 Carriage saddle. The carriage saddle shall consist of a bridge spaning the lathe bed, aligned by the bed ways, and designed to support all components of the carriage. The saddle shall be accurately finished and fitted to the bed ways. The saddle shall be provided with an adjustable gib or gibs and a locking arrangement for locking the carriage at any point on the lathe bed. Saddle wings shall be provided with wipers to prevent entry of chips and dirt between the carriage and bed ways.

3.6.4.3 Cross slide. The cross slide shall be fitted to the carriage saddle by means of dovetail construction. A full length adjustable gib shall be provided to take up wear. Design shall incorporate a guard to prevent the entry of chips and other foreign matter into the cross feed screw threads.

3.6.4.4 Compound rest. The compound rest shall consist of a base and a top, dovetailed together, and provided with an adjustable gib to take up wear. The compound rest shall be attached to the cross slide by means of an accurately fitted swivel bearing and shall be provided with a device for locking the compound rest at any point without pitting or deforming the cross slide mechanism. The compound shall be graduated from zero, as aligned at right angles to the bedways, in increments of one degree up to, and including, 60 degrees in both directions from zero. The compound rest shall be capable of being swiveled to any angle. The compound shall incorporate a T-slot for mounting a tool post.

3.6.4.5 Slide screws. The cross feed slide and angular feed slide screws shall be actuated by ball crank handles for hand operation of cross feed and angular feed. The screws shall be fitted with micrometer collars, graduated in one-thousandths of an inch, and capable of being zeroed at any point. A means shall be provided to enable locking the collars without pitting or marring the screws.

3.6.4.6 Apron. The apron shall be of one piece construction, designed to give support to all gear shafts and studs. Gearing in the apron shall provide power longitudinal and cross feeds. Controls shall be provided on the front of the apron for obtaining either right hand or left hand longitudinal power feeds and power cross feeds. Thread cutting feeds shall be obtained by means of split nut engagement of the lead screw. Engagement shall be effected by a control on the front of the apron. An automatic device shall be incorporated to prevent engagement of more than one feed at a time. A hand wheel mounted on the front of the apron shall provide longitudinal hand-feed. At the manufacturer's option, the reverse lever may be located convenient to the operator at the headstock end of the lathe.

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3.6.5 Lead screw. The lead screw shall have a major diameter of not less than 3/4 inch. Lead screw threading shall be Acme or modified square form.

3.6.6 Feed mechanism. Changes of feed shall be obtained by means of a quick change gear box mounted at the headstock end of the lathe bed. An index plate on the front of the gear box shall indicate the settings for different feeds.

3.6.6.1 Gear box. The gear box shall be enclosed at the top, front, and sides to keep out dirt and chips. It shall contain all gears necessary to cover the threading and feed ranges specified in Table I, with the exception that coarse threads (4 to 7 inclusive) and 27 pitch may be obtained by the addition or removal of pick-off gears. When this alternative is exercised, the necessary gears shall be furnished with each lathe.

3.6.7 Drive assembly. The drive assembly shall consist of all necessary motor brackets, countershaft brackets, pulleys, belts, tension adjustments, and any other equipment necessary to provide a complete drive assembly. The motor and countershaft shall be mounted on a frame or bracket either above the headstock, horizontally to the rear of the headstock, or beneath the headstock.

3.6.8 Electrical equipment.

3.6.8.1 Motor. Unless otherwise specified (see 6.2.1), the motor shall be not less than 1/2 horsepower (hp), 115 volt, single phase, 60 hertz (Hz), squirrel-cage induction-type with continuous duty rating in accordance with the requirements of CC-M-1807.

3.6.8.2 Motor control. A motor control switch with three positions (forward, stop, reverse) and a general purpose type enclosure shall be mounted in a convenient location on or near the headstock of the lathe.

3.6.8.3 Electrical components. Electrical components and equipment shall conform to NEMA ICS 1 and NFPA 79, as applicable. A start-stop, overload protected switch shall be provided for controlling the machine. No component shall be used in any application where the requirements exceed the electrical component manufacturer's rating for the part.

3.6.8.4 Power supply cord. An electrical power supply cord shall be installed on each lathe. Unless otherwise specified (see 6.2.1) the cord shall be not less than ten feet in length, wired into the lathe for operation, and equipped at the free end with a three prong grounding type plug. The cord shall be 120 volt, single phase, three conductor type meeting the requirements of NFPA 70 for sheathing, voltage, and amperage load carrying capacity as required by the lathe.

3.6.8.5 Color coded electrical circuitry. The wiring in the electrical system shall have individual point-to-point color coded wires in accordance with System II requirements of MIL-STD-681. The electrical system diagram/schematic furnished with each lathe shall be an exact duplicate of the electrical system and wiring in the lathe. The diagram/schematic shall positively identify the individual point-to-point color coded wires and wire termination points in the electrical system.

3.7 Standard equipment. The following standard equipment shall be furnished with each lathe.

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3.7.1 Wrench set. One set of all wrenches necessary to set up any operation specified herein.

3.7.2 Centers. Two lathe centers, hardened and ground, with tapers as specified in Table I. Tapers shall conform to ANSI B5.10-72.

3.7.3 Headstock spindle sleeve. One headstock spindle sleeve, ground, with tapers in accordance with ANSI B5.10-72. The outside taper shall be identical with the inside taper of the headstock spindle. The inside taper shall receive the lathe center.

3.7.4 Face plate. One face plate, 5 inches ± 1/2 inch in diameter, which shall be accurately fitted to the lathe spindle. The plate shall have one or more driving slots.

3.7.5 Tool post. One hardened tool post assembly suitable for mounting to the compound rest T-slot.

3.8 Accessories. When specified (see 6.2.1), the following accessories shall be furnished with each lathe.

3.8.1 Drill chuck. One drill chuck, three jaw, geared sleeve and pinion type, 0 to 1/2 inch capacity, with Morse No. 2 taper arbor in accordance with ANSI B5.10-72.

3.8.2 Independent chuck. One six inch, independent, four-jaw chuck fitted accurately to the lathe spindle. Jaws of the chuck shall be individually adjustable. The center hole shall be not less than 3/4 inch diameter. The chuck shall conform to the requirements of ANSI B5.8-72.

3.8.3 Universal chuck. One five inch, self centering, universal three-jaw chuck fitted accurately to the spindle of the lathe. The chuck shall have either one set of reversible jaws or two sets of jaws, one set capable of holding inside diameter work and the other set capable of holding outside diameter work. The center hole shall be not less than 3/4 inch diameter. The chuck shall conform to the requirements of ANSI B5.8-72.

3.8.4 Face plate. One large face plate, 7 inches ± 1/2 inch in diameter, fitted accurately to the lathe spindle. The plate shall have four or more equally spaced T-slots, or through slots, for clamping work or fixtures to the plate.

3.8.5 Steady rest. One steady rest of rigid construction, fitted accurately to the ways of the lathe. The steady rest shall be of the hinged type, or open sided with three-point bearing, with three adjustable jaws. Maximum capacity shall be not less than 2 1/2 inches.

3.8.6 Collet chuck and collets. One collet chuck of the hand wheel, draw-in type with a hollow draw bar. Collets shall be supplied with capacities, graduated in 1/64 inch, from 1/16 to 1/2 inch.

3.8.7 Dogs. One set of lathe dogs consisting of one of each of the following capacities:

3/8 inch
3/4 inch

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1 1/4 inch
1 1/2 inch

3.8.8 Tool holders. One set of hardened tool holders consisting of one of each of the following:

Boring bar holder
Boring bar, 1/2 inch diameter, seven inch length, 3/16 by 3/16 inch bit
Cut-off, left, with 3/32 inch thickness blade
Cut-off, right, with 3/32 inch thickness blade
Cut-off, straight, with 3/32 inch thickness blade
Turning, left, with 1/4 by 1/4 inch bit
Turning, right, with 1/4 by 1/4 inch bit
Turning, straight, with 1/4 by 1/4 inch bit
Knurling tool holder

3.8.9 Cutter bits. One set of hardened cutter bits as follows:

Item	Quantity
Bit, boring bar, 3/16 by 3/16 inch	5
Bit, turning, 1/4 by 1/4 inch	12
Blade, cut-off, 3/32 inch thickness	5

3.8.10 Miscellaneous accessories. The following miscellaneous accessories:

One pair fine diamond knurls
One pair medium diamond knurls
One pair coarse diamond knurls
One micrometer carriage stop
One set of metric transposing gears

3.9 Product identification. The lathe shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the lathe shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (f) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Operational test (see 4.6.1).
(c) Turning test (see 4.6.2).
(d) Facing test (see 4.6.3).
(e) Alignment accuracy test (see 4.6.5).
(f) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	1.0
Packaging inspection	S-4	4.0
Operational test	II	1.0
Turning test	II	1.0
Facing test	II	1.0
Alignment accuracy test	II	1.0

NOTE: Lot size shall be no more than 100 lathes.

4.5 Product Examination. Visually and manually examine the lathe to determine conformance with the requirements of 3.2 through 3.4.7 and 3.5.2 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of

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markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the lathe has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Operational test. The lathe shall be operated at no-load for not less than 30 minutes. Proper operation of all controls, meters, adjusting mechanisms, and other accessories shall be verified.

4.6.2 Turning test. To determine the turning accuracy of the lathe, a piece of SAE 1020 or 1040 cold rolled steel, 1 1/2 inches in diameter by 20 inches long, shall be used. The specimen should be turned to 1 7/16 inch ±0.002 inch diameter over a length of 18 inches. The allowable departure from a cylindrical surface shall be not greater than ±0.002 inch. The surface roughness shall be not greater than 63 microinches, aa, with no chatter marks (see 3.5.1).

4.6.3 Facing test. The facing test shall be made with SAE 1020 or 1040 cold rolled steel, approximately five inches in diameter by four inches long, chucked in a four-jaw chuck. Turn the outside diameter to 4 3/4 inches and face off the end. Surface roughness shall not be greater than 63 microinches, aa, and the faced surface shall be square with the cylindrical surface within 0.001 inch (see 3.5.1).

4.6.4 Thread cutting test. Using pieces of SAE 1020 or 1040 cold rolled steel, approximately 3/4 inches in diameter by ten inches long, cut a thread two inches long by 10 threads per inch. Undercut 1/4 inch and cut another thread two inches long by 14 threads per inch, then cut 4 threads, 18 threads, 24 threads, 32 threads, and 96 threads per inch and inspect for pitch. The threads shall be cut and tested both right hand and left hand as specified in Table I (see 3.5.2).

4.6.5 Alignment accuracy test. The lathe shall be tested in accordance with the standard test procedure of the National Machine Tool Builders Association and the illustrations in ANSI B5.16-72 to determine compliance with the accuracy requirements of 3.2. The center alignment test shall be performed using a mandrel approximately 1 1/2 inches in diameter by 12 inches long, set between centers. Indicate the headstock end by clamping an indicator in the lathe tool post and then traverse the carriage to the tailstock end and take a second reading. Vertical alignment shall be between zero and 0.001 inch high at the tailstock end. Horizontal alignment shall be within ±0.0005 inch (see 3.6.3.4).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A or B packing, and military marking shall be in accordance with MIL-M-18058.

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5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. This specification covers a motor-driven engine lathe for general use in performing any normal engine lathe operations within its capacity.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.7).
(e) Painting and finishing if different (see 3.4.7).
(f) Dimensions and characteristics if different (see 3.5.2).
(g) Motor if different (see 3.6.8.1).
(h) Power supply cord if different (see 3.6.8.4).
(i) Accessories when required (see 3.8).
(j) Product identification if different (see 3.9).
(k) First article inspection if different (see 4.3).
(l) Quality conformance inspection if different (see 4.4).
(m) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The lathe manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.3) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-11285 Rev D	Lathe, Engine, 9 By 22 Inch Capacity, Bench Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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